



University of Asia Pacific

Admit Card

Mid-Term Examination of Spring, 2026

Financial Clearance

PAID

Registration No : 22201243

Student Name : H. M. Tahsin Sheikh

Program : Bachelor of Science in Computer Science and Engineering



SLNO.	COURSE CODE	COURSE TITLE	CR.HR.	EXAM. SCHEDULE
1	CSE 404	Industrial Training	1.50	
2	CSE 400-B	Final Year Design Project	3.00	
3	BNG 401	Functional Bengali Language	3.00	
4	CSE 433	Blockchain and Distributed Security	3.00	
5	CSE 449	Design and Testing of VLSI	3.00	
6	CSE 434	Blockchain and Distributed Security Lab	1.50	
7	CSE 450	Design and Testing of VLSI Lab	1.50	

Total Credit: 16.50

1. Examinees are not allowed to enter the examination hall after 30 minutes of commencement of examination for mid semester examinations and 60 minutes for semester final examinations.

2. No examinees shall be allowed to submit their answer scripts before 50% of the allocated time of examination has elapsed.

3. No examinees would be allowed to go to washroom within the first 60 minutes of final examinations.

4. No student will be allowed to carry any books, bags, extra paper or cellular phone or objectionable items/incriminating paper in the examination hall. Violators will be subjects to disciplinary action.

This is a system generated Admit Card. No signature is required.

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 26
Program: B.Sc. in CSE
4th year 2nd Semester

Course Title: Blockchain and Distributed Security
Time: 1 hour. Credit Hour: 3.00

Course Code: CSE 433
Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

A company uses a blockchain-based payment system. A user creates a transaction and broadcasts it to the network. The transaction is received by several nodes and remains unconfirmed until a miner includes it in a block. Two different miners then solve the *PoW* puzzle at nearly the same time and broadcast their blocks. As a result, different nodes temporarily have different versions of the blockchain.

The company also plans to purchase 10 laptops from a company called "AoT" through an online marketplace that operates on a blockchain-based platform. The platform uses a smart contract to automatically manage the transaction between the buyer and seller when the required conditions are satisfied.

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|----|---|-----|-----------------|
| a. | Explain how the transaction moves through the blockchain network from its creation until it is included in a block. Mention the roles of the mempool, full nodes, and miners. | [3] | CLO1/
PLO-a |
| b. | Explain why the two miners can produce different valid blocks at approximately the same time and describe how the fork is resolved using the longest-chain rule. | [3] | CLO1 /
PLO-a |
| c. | For the purchase of the 10 laptops, write a step-by-step smart contract flow explaining and analyzing the conditions that must be satisfied and the actions automatically performed by the smart contract to complete the transaction between the company and the seller. | [4] | CLO3/
PLO-b |

QUESTION 2 [10 MARKS]

A group of eight organizations wants to establish a shared blockchain for managing an inter-organizational supply chain. The organizations do not want the general public to participate as validators. Each organization should maintain a copy of the ledger, and the system should continue operating even if one or more participating organizations fail. Since the organizations are known to each other, they want a consensus mechanism that is faster than *PoW*.

- | | | | |
|----|--|-----|----------------|
| a. | Identify whether this system is better described as a centralized, distributed, or decentralized system and explain your answer is based on control, ownership, and decision-making. | [2] | CLO1/
PLO-a |
| b. | Justify which type of blockchain—public, private, consortium, or hybrid—would be most appropriate for this application? Give two reasons based on the participation and ownership requirements. | [2] | CLO4/
PLO-c |
| c. | The organizations are considering <i>Raft</i> , <i>PBFT</i> , and <i>PoA</i> . Analyze which consensus mechanism would you recommend if some organizations may behave maliciously? | [3] | CLO3/
PLO-b |
| d. | To discourage dishonest behavior, the consortium introduces <i>PoS</i> , where validators lock cryptocurrency as stake. Now, a validator holding 30% of the total stake is selected to propose a block. Explain whether the selected validator guarantees that the block will be finalized and what must happen for the block to become finalized in a Proof-of-Stake (PoS) consensus mechanism? | [3] | CLO1/
PLO-a |

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 26
Program: B.Sc. in CSE
4th year 2nd Semester

Course Title: Design and Testing of VLSI
 Time: 1 hour. Credit Hour: 3.00

Course Code: CSE 449
 Full Marks: 20

There are two questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Explain the concept of circuit integration and how Moore's Law has influenced the design and development of modern smartphones and computers. [2] CLO1/PLOa

- b. [4] CLO2/PLOc

Table 1. Truth table of a circuit

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Analyze the given truth table in **Table 1** and determine the corresponding CMOS logic structure. Draw the CMOS circuit based on your analysis.

- c. [4] CLO2/PLOc

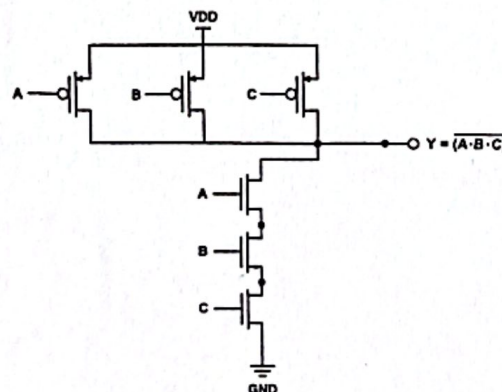


Figure 1. CMOS circuit for 3 input NAND

Apply CMOS layout design principles to draw the stick diagram of the 3-input NAND circuit shown in **Figure 1**.

QUESTION 2 [10 MARKS]

- a. Explain how a pass transistor transmits logic 0 and logic 1, and describe the limitations associated with transmitting each logic level. [2] CLO3/PLOd
- b. Examine how the lithography and etching processes work together to selectively define the regions required for CMOS transistor fabrication. [2] CLO1/PLOa
- c. An nMOS transistor is fabricated with a substrate doping concentration of $N_A = 8 \times 10^{17} \text{ cm}^{-3}$. Trapped charge density in oxide layer is $N_t = 4 \times 10^{10} \text{ cm}^{-2}$. The Fermi potential of the polysilicon gate $\phi_{\text{gate}} = -0.55 \text{ V}$. The transistor has an oxide thickness of $t_{\text{ox}} = 10 \text{ nm}$, and the gate is made of n^+ polysilicon. Assume temperature $T = 300 \text{ K}$, vacuum permittivity $\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$, relative permittivity $\epsilon_r = 11.7$, Boltzmann Constant $K = 1.38 \times 10^{-23} \text{ JK}^{-1}$, and intrinsic carrier concentration $N_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. A designer wants the transistor to have a threshold voltage of $V_{\text{TH}} = 0.70 \text{ V}$ under the condition source-bulk voltage $V_{\text{sb}} = 0$. Calculate the threshold voltage of the transistor using the appropriate NMOS threshold-voltage equation and determine whether the given device configuration will produce the required threshold voltage. Justify your answer. [6] CLO3/PLOd

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 26
Program: B.Sc. in CSE
4th year 2nd Semester

Course Title: Functional Bengali Language
Time: 1 hour.

Credit Hour: 3.00

Course Code: BNG 401
Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Explain the major components of a Natural Language Processing (NLP) pipeline. [3] CLO(a)/
Briefly describe how Bengali text processing introduces additional challenges PLO(a)
compared to English text processing.
- b. A Bengali online news platform wants to develop a Bangla text analysis system [7] CLO(a)/
to understand user comments and feedback. However, the collected PLO(b)
user-generated text contains several inconsistencies such as extra spaces, repeated
punctuation, and unstructured word boundaries. Before applying any NLP task,
the text needs to be converted into a clean and structured format.

text = " আজকের খবরটি অনেক ভালো!!! আমি বাংলা NLP শিখছি। ". As an
NLP engineer, you are asked to develop a basic preprocessing module for this
system. Write two Python functions:

1. normalize_text(text)

The function should:

- remove unnecessary leading and trailing spaces,
- replace multiple spaces with a single space,
- normalize repeated punctuation marks.

2. tokenize_text(text)

The function should:

- split the normalized text into meaningful tokens,
- separate punctuation marks from words.

QUESTION 2 [10 MARKS]

- a. Explain how understanding different levels of linguistic analysis can improve the performance of Bangla NLP systems. Discuss why considering the layered structure of language is important for processing Bengali text. [3] CLO(a)/ PLO(a)
- b. A Bengali document search system is being developed to retrieve relevant articles based on user queries. However, the system treats different grammatical forms of the same word as separate words, reducing the accuracy of search results. [7] CLO(b)/ PLO(b)

For example:

words = ["বইগুলো", "করছি", "করেছে", "ভালোবাসি"]

To solve this problem, the NLP engineer needs to implement a basic morphological normalization module.

Write Python functions to:

1. Implement a simple rule-based stemming function that removes common Bengali suffixes and returns the stem of each word.
2. Implement a simple lemmatization function using a dictionary-based lookup approach that maps Bengali word forms to their canonical forms.